

MECH 191 — Unit Quizzes

Applied, scenario-based short-answer quizzes, 5 questions per unit — each question requires using a formula, chart, or diagram from the course reference materials to reach an answer

MECH 191 — Metallurgy

Leeward Community College · Math and Science Division · Mechatronics Program

Part 1: Student Quiz (no answers) — Part 2: Instructor Answer Key

How to use these quizzes

Each unit has 5 short-answer questions built around a hypothetical shop or field scenario. Every question requires the student to actually apply a formula, chart, or diagram from the course reference materials — not just recall a definition — to reach their answer (e.g. calculating a carbon equivalent, reading an S-N curve, applying the galvanic series, interpreting a weld symbol).

Unit 1 — Materials Fundamentals

1. Using the Hall-Petch relationship $\sigma_y = \sigma_0 + k/\sqrt{d}$, a steel has $\sigma_0 = 70$ MPa and $k = 18$ MPa $\cdot$ mm^{0.5}. Calculate the yield strength for a fine-grained sample ($d = 0.005$ mm) and a coarse-grained sample ($d = 0.05$ mm). Which grain size would you specify for a high-strength structural bracket, and why?

Model answer. Fine grain ($d = 0.005$ mm): $\sigma_y = 70 + 18/\sqrt{0.005} = 70 + 254.6 \approx 325$ MPa. Coarse grain ($d = 0.05$ mm): $\sigma_y = 70 + 18/\sqrt{0.05} = 70 + 80.5 \approx 150$ MPa. The fine-grained sample is more than twice as strong, because more grain boundaries block dislocation motion — specify the fine-grained material for the high-strength bracket.

2. A tensile test on a steel specimen (original gauge length 50 mm) produces this data: the stress-strain curve becomes non-linear at 350 MPa, reaches a maximum stress of 550 MPa, and the specimen's final gauge length after fracture is 61 mm. Identify the yield strength, the ultimate tensile strength, and the % elongation at fracture. Is this steel ductile or brittle, and what visual feature would you expect on the fracture surface?

Model answer. Yield strength ≈ 350 MPa (onset of non-linearity). UTS = 550 MPa (maximum stress). % elongation = $(61-50)/50 \times 100 = 22\%$. With 22% elongation and a clear UTS-then-fracture curve, this steel is ductile — expect visible necking near the fracture and a cup-and-cone fracture surface, not a flat brittle break.

3. Referencing the packing factors of common crystal structures (SC $\sim 52\%$, BCC $\sim 68\%$, FCC $\sim 74\%$, HCP $\sim 74\%$) and their typical slip behavior, a coworker claims a BCC metal will cold-form into a complex shape just as easily as an FCC metal. Is your coworker correct? Justify your answer using the structures' packing and slip-system differences.

Model answer. No. FCC metals have more independent close-packed slip systems available at room temperature than BCC metals, which is why FCC metals (aluminum, copper, austenitic stainless) are generally more ductile and form more easily than BCC metals (ferritic steel, chromium) at room temperature — even though BCC's packing factor ($\sim 68\%$) is not far below FCC's ($\sim 74\%$), the difference in active slip systems is what drives the formability gap.

4. A quality engineer sectioning a failed part finds the crack initiated at a subsurface void roughly 0.5 mm across. Referencing the defect categories (point, line, planar, volume), which category does this void belong to, and how does its effect on strength differ from a single dislocation (a line defect) or a grain boundary (a planar defect)?

Model answer. A void is a volume (3D) defect. Unlike a single dislocation, which enables plastic deformation at the atomic scale, or a grain boundary, which blocks dislocation motion and can strengthen a material (Hall-Petch), a volume defect like a void removes load-bearing material altogether and acts as a stress concentrator — it is a much larger-scale flaw that directly reduces the effective cross-section and can initiate cracking well before the bulk material would otherwise fail.

5. An engineer must choose between a covalently-bonded ceramic and a metallically-bonded alloy for a bracket that will be dropped repeatedly during installation. Using what a metallic bond's "electron sea" allows compared to a ceramic's rigid, directional covalent bonds, which material is the safer choice for surviving impact, and why?

Model answer. The metallically-bonded alloy is the safer choice. Its delocalized "electron sea" allows atoms to shift position and re-bond continuously, so the metal absorbs impact energy through plastic deformation (bending, denting) rather than fracturing outright. A ceramic's rigid, directional covalent bonds cannot accommodate this kind of deformation, so it tends to shatter under the same impact instead of deforming.

Unit 2 — Steel & Alloys

1. A structural fabricator is welding an alloy steel plate with this certified composition: %C = 0.28, %Mn = 1.40, %Cr = 0.50, %Mo = 0.20, %V = 0.05, %Ni = 0.30, %Cu = 0.15. Using $CE = \%C + \%Mn/6 + (\%Cr + \%Mo + \%V)/5 + (\%Ni + \%Cu)/15$ and the preheat bands (CE < 0.40: none; 0.40-0.60: 100-200°C; > 0.60: 200-300°C+, mandatory), calculate the CE and state the required preheat.

Model answer. $CE = 0.28 + 1.40/6 + (0.50 + 0.20 + 0.05)/5 + (0.30 + 0.15)/15 = 0.28 + 0.233 + 0.150 + 0.030 \approx 0.69$. Since CE > 0.60, preheat of 200-300°C or higher is mandatory before welding.

2. A steel sample contains 0.40% carbon. Referencing the iron-carbon phase diagram and the eutectoid point (0.77% C), is this steel hypoeutectoid, eutectoid, or hypereutectoid? After slow cooling from austenite, would its room-temperature microstructure contain more ferrite or more pearlite?

Model answer. 0.40% C is below the eutectoid point (0.77% C), so this is a hypoeutectoid steel. On slow cooling it forms a mixture of ferrite and pearlite, and because 0.40% is well below 0.77%, the microstructure will be predominantly ferrite with a smaller fraction of pearlite.

3. A CCT diagram for an alloy steel shows: cooling at 50°C/s produces 100% martensite, cooling at 5°C/s produces a bainite/martensite mix, and cooling at 0.5°C/s produces pearlite. A technician air-cools a thick section of this steel and measures a core cooling rate of about 1°C/s. What microstructure would you predict at the core, and why might the thin outer surface of the same part end up with a different microstructure than the core?

Model answer. At roughly 1°C/s, the core would fall between the bainite/martensite curve (5°C/s) and the pearlite curve (0.5°C/s), so expect a mixed bainite/pearlite (or bainite-dominant) microstructure. The surface cools faster than the core because it loses heat directly to the surrounding air/quenchant while the core is insulated by the surrounding metal, so the surface may reach a higher cooling rate and form more martensite than the core — a common cause of a hard case with a softer interior in air-cooled or mildly quenched parts.

4. Using $PREN = \%Cr + 3.3 \times \%Mo + 16 \times \%N$, calculate the PREN for Grade A (18% Cr, 0% Mo, 0.05% N) and Grade B (17% Cr, 2.5% Mo, 0.15% N). Referencing the PREN bands (<18 low · 18-25 moderate/fresh water · 25-35 good/brackish-marine · 35-40 high/seawater · >40 very high), which grade would you specify for seawater piping?

Model answer. Grade A: $PREN = 18 + 0 + 0.8 = 18.8$ (moderate, fresh-water range). Grade B: $PREN = 17 + 8.25 + 2.4 = 27.65$ (good, brackish/marine range). Grade B is the clear choice between the two for a chloride-rich environment; note that for full open-seawater service, an even higher-PREN duplex or super-austenitic grade (PREN > 35) would be preferable to either option shown here.

5. A drawing calls out an aluminum bracket as "6061-T651." Using the Aluminum Association designation system, explain what the alloy number "6061" and the temper "T651" each tell you, and describe the sequence of processing steps the raw stock went through to reach this final condition.

Model answer. "6061" identifies the alloy family (6xxx series, magnesium-silicon alloyed, heat-treatable). "T651" means the material was solution heat-treated, artificially aged (T6), and then additionally stress-relieved by stretching (the "51" suffix) — a sequence used for plate that must machine flat without distorting. So the stock was solution treated, quenched, stretched to relieve residual stress, and then aged at an elevated temperature to reach its final strength.

Unit 3 — Testing, Inspection & Quality

1. A Brinell hardness test uses a 10 mm diameter ball under a 3000 kgf load, producing a 4.2 mm diameter indentation. Using $HB = 2F / [\pi D(D - \sqrt{D^2 - d^2})]$, calculate the Brinell hardness number, then use $UTS(MPa) \approx 3.45 \times HB$ to estimate this steel's ultimate tensile strength.

Model answer. $D^2 - d^2 = 100 - 17.64 = 82.36$, $\sqrt{82.36} \approx 9.075$. $D - 9.075 = 0.925$. $HB = (2 \times 3000) / (\pi \times 10 \times 0.925) = 6000 / 29.06 \approx 207$ HB. Estimated $UTS \approx 3.45 \times 207 \approx 710$ MPa.

2. An S-N curve for a steel alloy shows: at 400 MPa stress amplitude the part survives about 10^4 cycles, at 300 MPa about 10^6 cycles, and the curve flattens at 250 MPa (the endurance limit). A component in service sees a cyclic stress amplitude of 240 MPa. Based on this curve, would you expect the component to eventually fail from fatigue? Explain.

Model answer. No — 240 MPa is below the 250 MPa endurance limit shown on the S-N curve. For steels, stresses below the endurance limit are expected to allow essentially infinite cycles without fatigue failure, so this component should not be expected to fail from fatigue under this loading.

3. An impact-energy-vs-temperature curve for a pressure vessel steel shows 15 J absorbed at -40°C (lower shelf), 60 J at 0°C (transition region), and 120 J at 40°C (upper shelf). This vessel will operate in an environment that regularly reaches -30°C . What concern does this raise, and what does the curve tell you about the material's suitability at that temperature?

Model answer. At -30°C the material is likely still on or near the lower shelf (close to the -40°C value of only 15 J), meaning it would behave in a brittle manner — absorbing little energy and fracturing suddenly with little warning, well below its ductile (upper shelf) toughness. This is a serious concern for a pressure vessel; a tougher material with a lower DBTT, or a design change to keep the vessel above its transition temperature range, should be considered.

4. A weld inspector needs to detect a suspected lack-of-fusion defect located 15 mm below the surface of a 25 mm thick steel plate. Referencing the capabilities of VT, PT, MT, UT, and RT, which method(s) could actually detect this defect, and which should be ruled out immediately, and why?

Model answer. VT and PT should be ruled out immediately — both only detect defects open to or breaking the surface, and this defect is fully subsurface. MT can only detect defects at or very near the surface, so it would likely miss a defect 15 mm deep as well. UT and RT are the methods that can actually detect this defect, since both can locate volumetric/subsurface discontinuities well below the surface.

5. A magnetic particle test reveals a linear indication 3 mm long at the toe of a fillet weld. The governing code states that linear indications longer than 1.5 mm in this location are unacceptable. Walk through how this finding moves through the indication → discontinuity → defect → reject sequence, and state the final disposition of the weld.

Model answer. Indication: the MT test produces a visible response at the weld toe. Discontinuity: on evaluation, this is confirmed to be a real linear interruption in the material (not a false indication), measuring 3 mm. Defect: since the code's acceptance limit for this location is 1.5 mm and the actual discontinuity is 3 mm — double the limit — it is classified as a defect. Reject: the weld is rejected and must be repaired (or the section removed and rewelded) per the governing code.

Unit 4 — Manufacturing Processes

1. A part is turned with a feed of 0.3 mm/rev using a tool with a 0.8 mm nose radius. Using $Ra \approx f^2/(8r)$, calculate the theoretical surface roughness in μm . The print requires $Ra \leq 3.2 \mu\text{m}$. Is this parameter set acceptable, and if not, name two parameter changes that would fix it without changing the tool.

Model answer. $Ra = (0.3)^2 / (8 \times 0.8) = 0.09 / 6.4 = 0.0141 \text{ mm} = 14.1 \mu\text{m}$ — far rougher than the $3.2 \mu\text{m}$ requirement, so this parameter set is not acceptable. Reducing the feed rate and/or specifying a tool with a larger nose radius would both reduce Ra without needing to change cutting speed.

2. A radiograph of a casting shows an irregular, jagged void concentrated near the last region of the casting to solidify, rather than smooth, evenly distributed spherical voids. Referencing the difference between shrinkage porosity and gas porosity, which defect is this, and what change to the gating/risering system would you recommend?

Model answer. This is shrinkage porosity — its irregular shape and location at the last-to-solidify (thermal center) region are the signature of solidification shrinkage that wasn't fed with additional liquid metal. Recommend adding or enlarging a riser (feeder) positioned to stay liquid longest, so it can continue feeding molten metal into that region as it shrinks during solidification.

3. A shop needs to produce 500 identical brackets with a complex internal pocket and tight tolerances, with minimal per-part setup time after the first unit. Comparing sand casting, CNC milling, and powder metallurgy for this job, which would you recommend, and what's the key limiting factor ruling out the other two?

Model answer. CNC milling is the best fit here: sand casting requires a new mold/pattern setup with rough tolerances and can't easily produce a complex internal pocket without cores, and powder metallurgy requires an expensive die and is best justified at much higher volumes (often 10,000+ parts) to offset tooling cost — 500 parts doesn't reach that break-even point. CNC milling handles moderate volumes with tight tolerances and complex geometry without new tooling for each part.

4. An engineer compares a forged crankshaft to one machined from solid billet, both from the same alloy and final geometry. Referencing what happens to grain flow during forging versus machining, explain why the forged part is generally stronger along its load path.

Model answer. Forging deforms the metal's existing grain structure to follow the part's contours, producing continuous, unbroken grain flow that lines up with the load path and the part's shape. Machining from billet cuts through the original grain structure, exposing grain ends at the surface and along stress-critical features rather than following them — the forged part's aligned grain flow gives it better strength and fatigue resistance along the load path even with identical composition and geometry.

5. A powder metallurgy iron part is sintered at roughly 1100-1150°C — about 70-90% of iron's melting point, not fully melting the powder. Referencing the three sintering stages (burn-off, neck formation, densification/grain growth), explain why sintering works below the melting point rather than melting the part.

Model answer. Sintering relies on solid-state diffusion, not melting: after burn-off removes the pressing lubricant, atoms at the contact points between powder particles diffuse into each other at high temperature, forming necks that grow and gradually densify the part while it remains solid. Full melting would destroy the pressed geometry and require a completely different (casting) process — sintering's advantage is that it bonds the powder into a solid part while preserving the near-net shape achieved during compaction.

Unit 5 — Joining, Failure & Corrosion

1. A welder uses 25 V, 180 A, and a travel speed of 300 mm/min. Using $HI = (V \times A \times 60) / (1000 \times \text{travel speed})$, calculate the heat input in kJ/mm. If the joint's WPS specifies a maximum heat input of 1.5 kJ/mm to avoid excessive HAZ grain coarsening, is this parameter set acceptable?

Model answer. $HI = (25 \times 180 \times 60) / (1000 \times 300) = 270,000 / 300,000 = 0.9 \text{ kJ/mm}$. Since 0.9 kJ/mm is below the 1.5 kJ/mm maximum, this parameter set is acceptable.

2. A welding drawing shows a fillet weld symbol with "6" written to the left of the symbol on the arrow side of the reference line, and a tail containing the note "GMAW." Referencing weld symbol anatomy (reference line, arrow/other side, tail), what does the "6" specify, which side of the joint does the weld go on, and what does the tail note tell the welder?

Model answer. The "6" specifies the fillet weld's leg size (6 mm). Because the symbol is on the arrow side of the reference line, the weld goes on the side of the joint the arrow physically points to. The tail note ("GMAW") specifies the welding process to be used for this weld.

3. Referencing the galvanic series (most active/anodic like magnesium down to most noble/cathodic like gold/platinum, with steel and aluminum in between), an aluminum plate is fastened with stainless steel bolts in a marine environment. Which metal corrodes preferentially, and would the corrosion be worse with a small stainless bolt in a large aluminum plate, or a large stainless plate with small aluminum rivets? Explain using the area effect.

Model answer. Aluminum is more active (anodic) than stainless steel, so the aluminum corrodes preferentially. The area effect makes corrosion severity depend on the anode-to-cathode area ratio: a small anodic area (small aluminum rivets) paired with a large cathodic area (large stainless plate) concentrates all the galvanic current onto that small anodic area, causing severe, rapid local attack — this is the worse case. A small stainless bolt in a large aluminum plate spreads the same current over a much larger anodic area, causing much milder, more distributed corrosion.

4. A structural component fails unexpectedly in service. Walk through the standard failure analysis methodology in order (visual inspection → NDT → mechanical/metallographic testing → root cause → report), and explain why skipping straight to metallographic sectioning without a visual inspection first risks losing evidence needed to determine the failure's origin.

Model answer. The correct order is: visual inspection (document the as-found condition, fracture appearance, and surrounding context) → NDT (locate and characterize other defects without altering the part) → mechanical/metallographic testing (destructive sectioning and microstructural examination) → root cause determination → final report. Jumping straight to metallographic sectioning destroys the original fracture surface and part geometry before it has been documented or fully inspected non-destructively — features like beach marks, fracture origin, or corrosion patterns visible only on the intact surface can be permanently lost, making it much harder (or impossible) to reconstruct how and where the failure actually started.

5. A buried pipeline requires cathodic protection to a target of -0.85 V vs. Cu/CuSO₄ per NACE SP0169. Referencing sacrificial anode vs. impressed current systems, if the pipeline is very long (50+ km) and needs a large protective current, which system is generally more practical, and why might a sacrificial anode system struggle over that distance?

Model answer. Impressed current cathodic protection is generally more practical for a long pipeline needing high current, since it uses an external DC power source that can supply as much current as needed and can be adjusted over time. A sacrificial anode system relies purely on the natural driving voltage between the anode metal and the pipeline, which limits the current and protective range each anode can provide — protecting 50+ km would require an impractically large number of anodes, and the system offers no way to increase current output as demand changes or anodes deplete.